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AttriSelect: Efficient Attribute-Driven Data Selection for Targeted Vision Model Repair

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01 May 2026 (modified: 27 May 2026) NeurIPS 2026 Conference Submission
Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors
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TL;DR: Attribute-Driven Data-Centric Bug Discovery and Repair

Abstract:
Targeted vision repair aims to correct model failures on rare semantic slices after a debugging pipeline identifies the failure slice and a generator produces corresponding repair images. In most existing workflows, these repair images are appended to the original training set, and the model is retrained or fine-tuned using all source data. While this preserves broad coverage, it also makes each repair cycle computationally expensive and time-consuming, especially as repairs accumulate over multiple iterations. Moreover, it provides little control over which source examples actually contribute to the repair. This poses a data selection problem related to active learning: rather than using the full dataset, the goal is to identify a compact yet informative subset that best supports repair while substantially reducing retraining cost. However, an effective repair subset must satisfy requirements beyond standard sample efficiency. In particular, it should preserve class coverage, retain rare repair-relevant attributes, and avoid redundancy with the generated repair data. Existing active learning methods typically prioritize uncertainty, diversity, or gradient-space coverage, but these signals are often poorly aligned with the semantic attributes that define the failure slice. To address this limitation, we propose AttriSelect, an attribute-driven selector that prioritizes samples based on attribute rarity, generated-data coverage, and soft class balance. We provide theoretical analysis and evaluate AttriSelect across image classification, pose estimation, and object detection. Across tasks and repair budgets, AttriSelect gives the strongest repair results in most settings, showing that explicit attribute metadata provides a repair-specific selection signal distinct from traditional active learning criteria.

Checklist Confirmation: I confirm that I have included a paper checklist in the paper PDF.
Supplementary Material: zip (/attachment?id=JxkGmySuLb&name=supplementary_material)
Responsible Reviewing: We acknowledge the responsible reviewing obligations as authors.
Primary Area: Data-centric aspects of AI (e.g., data selection, active learning, curriculum learning, online learning)
Secondary Area: Computer vision (e.g., object detection, segmentation, image classification, image generation, video)
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Reviewer Nomination: Ouyang Xu (/profile?id=~Ouyang_Xu1)
Submission Number: 12417

Official Review of Submission12417 by Reviewer yyPZ

Official Review by Reviewer yyPZ 25 Jun 2026, 01:34 (modified: 23 Jul 2026, 09:35)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Revisions (/revisions?id=gFhG8tCROU)

Summary:
The paper introduces AttriSelect, a targeted data selection algorithm designed to efficiently repair semantic failure slices in computer vision models. Rather than relying on traditional active learning metrics like uncertainty, gradient embeddings, or geometric coverage, AttriSelect scores source training data using structural attribute metadata assigned by Vision-Language Models (VLMs).

The scoring function balances three components: class-conditional attribute rarity, a penalty for attribute values already abundant in the generated synthetic repair data, and a soft class-balancing term.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths:

1. Framing targeted model repair as an attribute-aware source selection problem is a highly practical perspective.
2. The methodology is straightforward.
3. Writing is clear.

Weakness:

1. Misleading Claims Regarding Computational Efficiency. The authors highlight in Table 2 that AttriSelect takes less than 2 seconds, claiming a speedup of up to 283× over baselines. However, this entirely omits the massive computational overhead of the prerequisite step: running Qwen3-VL to annotate the entire source pool. Appendix A states it takes 10–20 minutes to annotate just 1,000 images. For datasets like CelebA (80,000 images) or COCO (20,000 images), this requires tens to hundreds of GPU hours. Claiming high efficiency while ignoring the cost of generating the metadata that the algorithm relies upon is highly misleading.
2. Comparison look Unfair. AttriSelect is given privileged information like VLM-assigned attributes and explicit bug definitions, but the compared active learning baselines (Entropy, BADGE, CoreSet) do not have access to. The authors should compare it against a baseline with the same information, for example, an "Attribute-Filtered Random" or "Attribute-Filtered Entropy" baseline.
3. Despite the heavy pipeline (VLM tagging, diffusion generation, subset selection), the absolute Fix Rates (FR) in Table 1 remain quite low, mostly ranging between 5% and 13%.

Quality: 2: not good
Clarity: 3: good
Significance: 2: not good
Originality: 3: good
Questions:
See weaknesses

Limitations:
yes

Rating: 2: Reject: For instance, a paper with technical flaws, weak evaluation, inadequate reproducibility and incompletely addressed ethical considerations.
Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
NA

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Official Review of Submission12417 by Reviewer hG6j

Official Review by Reviewer hG6j 19 Jun 2026, 05:39 (modified: 23 Jul 2026, 09:35)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j
Revisions (/revisions?id=qvZIzZToTG)

Summary:
This paper studies efficient source-data selection for targeted vision model repair. Given an already diagnosed failure slice and a fixed set of generated repair images, the goal is to choose a compact subset of the original source data for retraining. The proposed method, AttriSelect, scores source examples using class-conditional attribute rarity, a generated-data frequency penalty, and soft class balancing. Experiments on classification, pose estimation, and detection show that AttriSelect often outperforms generic active learning baselines and can sometimes outperform full-source repair.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
Strength: The paper addresses a useful and relatively underexplored problem: after a bug slice has been found and repair images have been generated, which original source examples should be used for efficient repair? This is a practical question that is not well covered by standard active learning. The method is simple, intuitive, and well aligned with the problem. The three scoring terms each have a clear role: rarity targets rare semantic attributes, the generated-data penalty avoids redundancy with synthetic repair data, and soft class balancing avoids class collapse. The experiments are reasonably broad and include several strong active learning baselines. The runtime advantage is also convincing.

Weakness The main weakness is that the method depends heavily on the upstream slice discovery, generated data quality, and VLM-assigned attributes. AttriSelect does not itself discover the failure slice, and the paper could be clearer about how robust it is to different slice definitions or imperfect attribute vocabularies. The algorithmic novelty is also moderate: the score is useful but fairly straightforward. I would also like to see stronger attribute-aware baselines, such as selecting target-slice examples plus stratified background samples.

Quality: 3: good
Clarity: 3: good
Significance: 3: good
Originality: 3: good
Questions:
1. How sensitive is AttriSelect to the upstream slice discovery method and slice granularity?
2. Can the authors compare with simple attribute-aware baselines, such as exact target-slice selection plus stratified sampling?
3. What is the end-to-end cost including generation, VLM attribute assignment, selection, and retraining?
4. How does the method behave when the failure slice is not well captured by the closed-set attribute vocabulary?

Limitations:

The paper discusses attribute noise and includes a human audit, which is helpful. However, it should more explicitly discuss dependence on upstream slice discovery, VLM attribute quality, generator quality, and end-to-end repair cost.

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.
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Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
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Code Of Conduct Acknowledgement: Yes
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Official Review of Submission12417 by Reviewer KaLa

Official Review by Reviewer KaLa 15 Jun 2026, 03:38 (modified: 23 Jul 2026, 09:35)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer KaLa
Revisions (/revisions?id=12XbHqOFYG)

Summary:
This paper studies targeted vision model repair from the perspective of repair-time source-data selection. Given a diagnosed failure slice and generated repair images, the proposed AttriSelect method selects a compact subset of original training examples to accompany the generated data during retraining, aiming to reduce computational cost while preserving repair effectiveness. The method uses attribute-driven criteria, including class-conditional rarity, generated-data coverage control, and soft class balance, to better capture repair-relevant semantic slices than conventional uncertainty-, diversity-, or gradient-based active learning methods. Experiments across image classification, pose estimation, and object detection suggest the effectiveness of AttriSelect.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
Strengths.

1. The paper presents an interesting and inspiring perspective by formulating data selection as an important component of the targeted model repair process.
2. The experimental results are promising and suggest that the proposed attribute-driven selection strategy can improve repair performance across different vision tasks.

Weaknesses.

1. The problem formulation needs to be clarified. It is not always clear whether the proposed method is selecting generated repair data, original source data, or a combination of both. The descriptions in the abstract, introduction, and conclusion appear somewhat vague or potentially inconsistent on this point. The authors are encouraged to state explicitly what data pool AttriSelect operates on and how the selected data are used during retraining.
2. The categorization of prior model repair methods requires further clarification. In particular, reference [7] seems to be associated with both the first and second lines of work, which makes the taxonomy confusing. The authors should revise this discussion to make the distinctions among different categories more precise.
3. It is unclear what repair method or methods are used together with AttriSelect in the experiments. Since AttriSelect is a data selection strategy rather than a complete repair algorithm, the experimental section should clearly specify the underlying repair pipeline, retraining/fine-tuning procedure, and how the selected source samples are combined with the generated repair images.
4. The paper states that AttriSelect is best viewed as one component in a localized repair pipeline. However, the experiments do not appear to evaluate its compatibility with multiple repair pipelines. It would strengthen the paper if the authors could test AttriSelect with different repair methods or, at least, discuss the extent to which the conclusions depend on the specific repair pipeline used in the experiments.

5. The formatting of the references should be improved. For example, the names of journals and conferences should be made consistent, either using full names or standard abbreviations throughout the reference list.

Quality: 2: not good
Clarity: 2: not good
Significance: 3: good
Originality: 3: good
Questions:

1. Could the authors clearly specify whether AttriSelect selects from the original source/training data, the generated repair data, or both at related parts?
2. Since AttriSelect is presented as a data selection component rather than a complete repair method, could the authors specify the exact repair pipeline used in each experiment?
3. The paper states that AttriSelect is best viewed as one component in a localized repair pipeline. Could the authors provide evidence or discussion about whether AttriSelect remains effective when paired with different repair pipelines or retraining strategies?
4. Could the authors improve the categorization of prior repair methods?

Limitations:
yes

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
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